Leveraging Out-of-Distribution Data for Clinically Meaningful Predictions

Francisco Haas | University of Michigan College of Engineering
Mentor: Associate Professor Jenna Wiens

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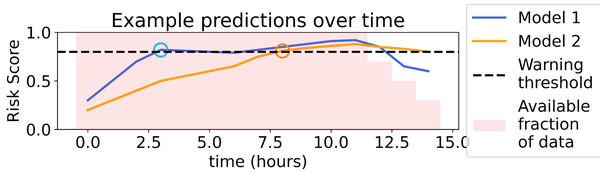
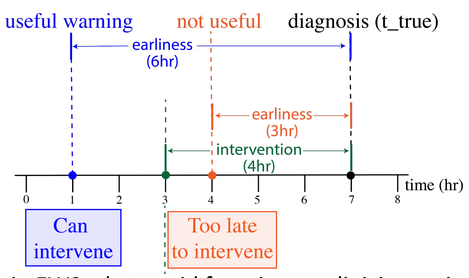
Introduction

Sepsis is the #1 cause of inhospital mortality

Early warning systems (EWSs) can estimate deterioration risk over time; first time risk exceeds a threshold → warning time



Motivation: EWSs are most useful if they contribute novel insights that can inform intervention.[1]



Goal: Train EWSs that avoid focusing on clinician actions over other variables ⟶ earlier predictions.

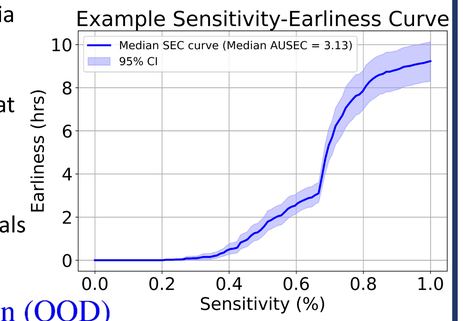
Methods

Earliness: time from model warning to when a patient meets outcome criteria

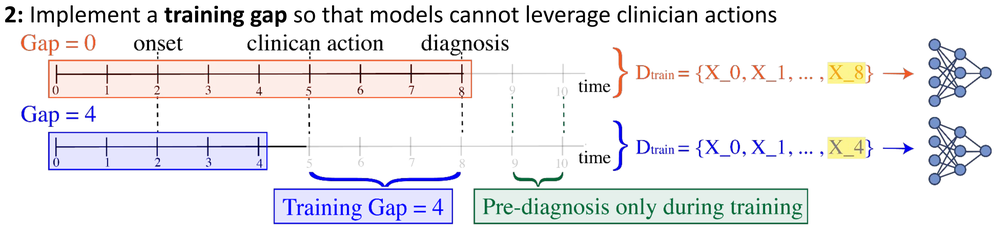
- For comparison across sensitivity thresholds, we use the **Area Under the Sensitivity-Earliness Curve (AUSEC)**
- AUSEC is interpreted as the expected amount of time before diagnosis that an EWS will sound an alarm across all levels of sensitivity

Area Under the Receiver Operating Characteristic (AUROC) is used as the standard measure of discriminative performance.

Approach 1: Train models **out-of-distribution** with data from multiple hospitals so that models learn to leverage stable medical variables, rather than care processes which vary between hospitals



Approach 2: Implement a **training gap** so that models cannot leverage clinician actions



For both approaches, we expect a tradeoff between AUSEC and AUROC: **As models make earlier predictions (AUSEC increases), discriminative performance decreases (AUROC decreases)**

Results

Synthetic Data (In-distribution vs OOD)

Table 1: Training OOD resulted in significant improvement in earliness for only a small reduction in discriminative performance

Median Metric (1000 Bootstraps)	Model 1: In-distribution	Model 2: Trained OOD
AUROC (95% CI)	**0.823** (0.818, 0.827)	**0.816** (0.812, 0.821)
AUSEC (hrs) (95% CI)	**3.13** (2.81, 3.46)	**3.76** (3.36, 4.17)

Table 2: Training OOD placed less importance on clinician action (and decreased variability) while importance on other variables remained approximately constant

Median Abs. Coefficient	Model 1: In-distribution	Model 2: Trained OOD
Clinician Action Coefficient	**0.256** (0.077, 0.409)	**0.212** (0.114, 0.253)
Other Variable Coefficients	**0.137** (0.131, 0.148)	**0.138** (0.126, 0.156)

Real Data (Training Gap)

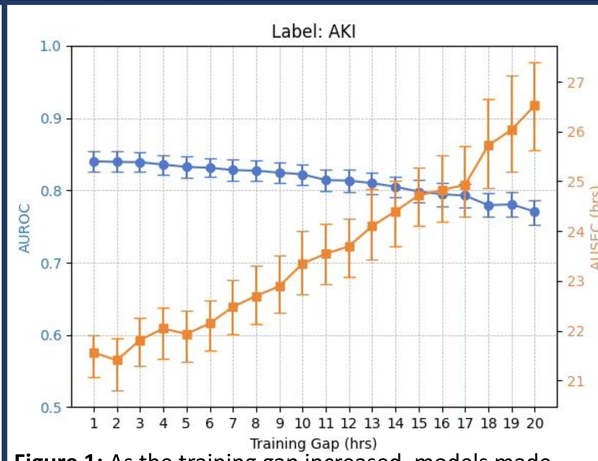


Figure 1: As the training gap increased, models made earlier predictions while discriminative performance (AUROC) decreased slightly

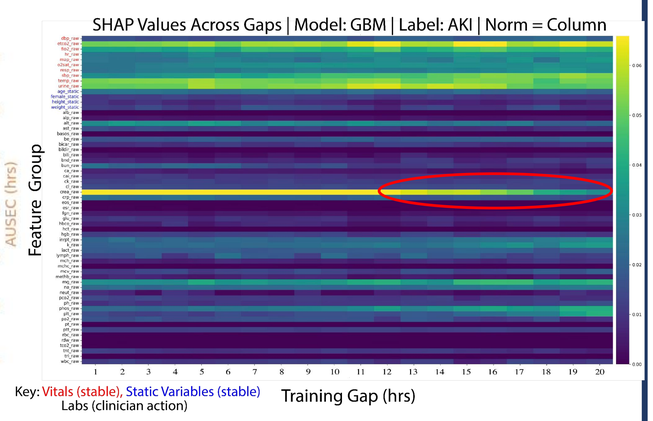


Figure 2: As the training gap increased, model importance on the creatinine lab (ordered when clinicians suspect AKI) decreases, while importance on vitals increase slightly

Conclusion: both approaches resulted in improvements in earliness with moderate decreases in discriminative performance, suggesting they can be used to improve the clinical utility of EWSs.

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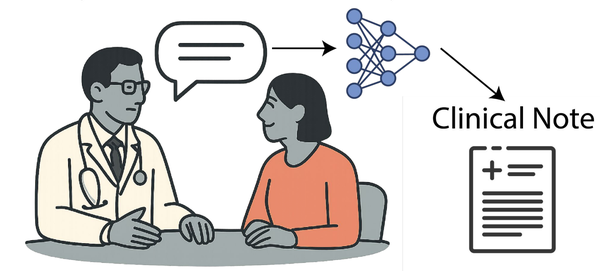
Side Project: Ambient AI

Recently, **Ambient AI** has been implemented in healthcare settings to aid clinicians with clinical note generation. These tools have been shown to correlate with **decreased time in notes per appointment, mental demand on clinicians**, and other benefits.[2]



While the benefits for clinicians are evident, **how has the use of Ambient AI at Michigan Medicine changed the clinical notes that patients see?**

AI Clinical Notes at Michigan Medicine

Method: Matched case-control study comparing clinical notes for the same patient, clinician, and visit type before and after ambient AI adoption, allowing for direct analysis of changes in note content and structure attributable to the AI system.

Results:

Table 3: With the use of Ambient AI at Michigan Medicine, clinical notes have become longer, more complex, and more repetitive, suggesting notes become harder to read and less informative

Metrics Mean (Std)	Pre AI (n = 233,223)	Post AI (n = 187,004)
Length	6179.6	6454.1
Word Count	801	802
Mean Word Length	5.23	5.34
Measure of Textual Lexical Diversity (MLTD)[3] (lower is more repetitive)	160.11 (94.13)	146.73 (75.41)
Readability: Flesch-Kincaid (FK) grade level[4] (higher is harder to read)	10.52 (3.68)	11.60 (5.20)

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